

Task 6: Session 6 (AIML) - Assignment Questions

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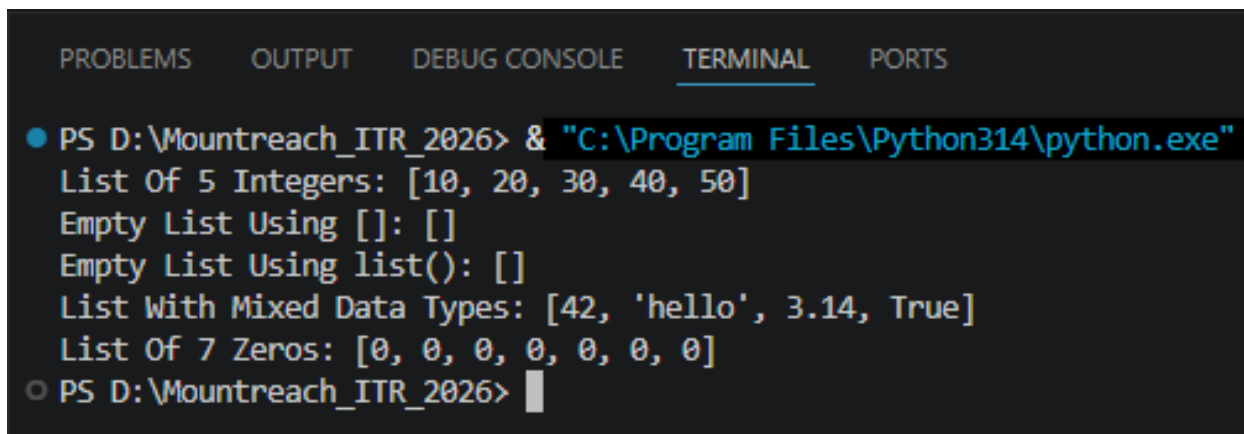
1. Creating Lists

Write a program to create the following lists using different methods:

- A list of 5 integers directly.
- An empty list using both [] and list().
- A list containing mixed data types (int, string, float, boolean).
- A list of 7 zeros using repetition (*).

Print all lists with clear labels and add comments explaining each creation method.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
List Of 5 Integers: [10, 20, 30, 40, 50]
Empty List Using []: []
Empty List Using list(): []
List With Mixed Data Types: [42, 'hello', 3.14, True]
List Of 7 Zeros: [0, 0, 0, 0, 0, 0, 0]
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
# A List Of 5 Integers Created Directly Using Square Brackets
integers = [10, 20, 30, 40, 50]
print("List Of 5 Integers:", integers)

# Empty List Using [] & list()
empty_list_1 = []
empty_list_2 = list()
```

```
print("Empty List Using []:", empty_list_1)
print("Empty List Using list():", empty_list_2)

# A List With Mixed Data Types
mixed = [42, "hello", 3.14, True]
print("List With Mixed Data Types:", mixed)

# A List Of 7 Zeros Using Repetition Operator (*)
zeros = [0] * 7
print("List Of 7 Zeros:", zeros)
```

2. Indexing and Negative Indexing

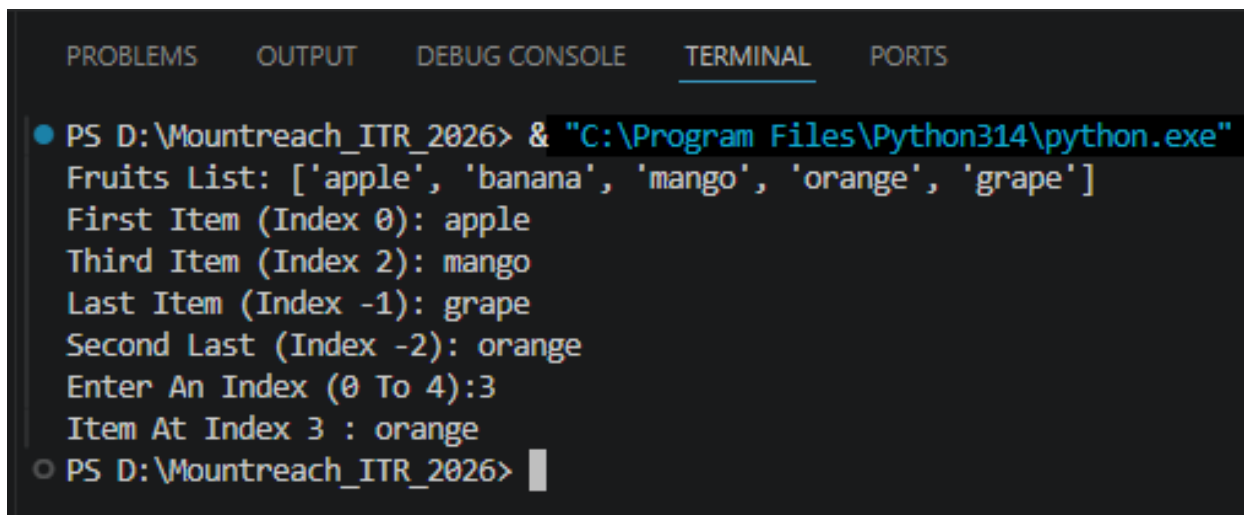
Create a list: `fruits = ["apple", "banana", "mango", "orange", "grape"]`

Write code to print:

- First item
- Third item
- Last item (using negative index)
- Second last item (using negative index)

Take a number as input from the user and print the item at that index (handle basic cases only).

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Fruits List: ['apple', 'banana', 'mango', 'orange', 'grape']
First Item (Index 0): apple
Third Item (Index 2): mango
Last Item (Index -1): grape
Second Last (Index -2): orange
Enter An Index (0 To 4):3
Item At Index 3 : orange
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
fruits = ["apple", "banana", "mango", "orange", "grape"]
print("Fruits List:", fruits)

print("First Item (Index 0):", fruits[0])
print("Third Item (Index 2):", fruits[2])
print("Last Item (Index -1):", fruits[-1])
print("Second Last (Index -2):", fruits[-2])

i = int(input("Enter An Index (0 To 4):"))
print("Item At Index", i, ":", fruits[i])
```

3. Slicing Lists

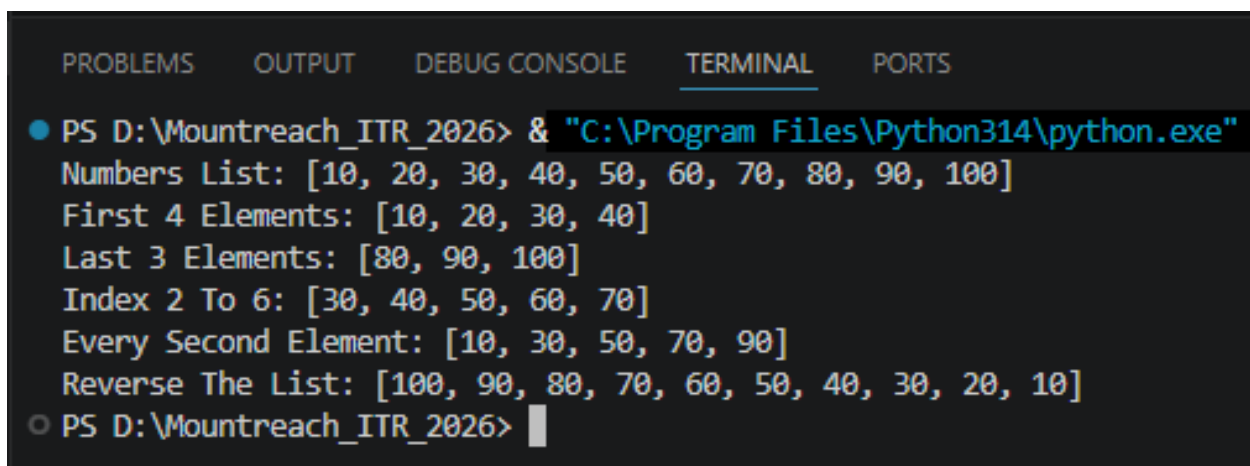
Given the list: numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

Using slicing, print:

- First 4 elements
- Last 3 elements
- Elements from index 2 to 7
- Every second element
- The entire list in reverse order

Add comments explaining slicing syntax.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Numbers List: [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
First 4 Elements: [10, 20, 30, 40]
Last 3 Elements: [80, 90, 100]
Index 2 To 6: [30, 40, 50, 60, 70]
Every Second Element: [10, 30, 50, 70, 90]
Reverse The List: [100, 90, 80, 70, 60, 50, 40, 30, 20, 10]
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
print("Numbers List:", numbers)

# Syntax: list[start:stop:step] (stop Is Exclusive)

# First 4 Elements -> start=0 (default), stop=4
print("First 4 Elements:", numbers[:4])

# Last 3 Elements -> start=-3, stop=end (default)
print("Last 3 Elements:", numbers[-3:])

# Elements From Index 2 To 7 -> start=2, stop=7
print("Index 2 To 6:", numbers[2:7])
```

```
# Every Second Element -> step=2
print("Every Second Element:", numbers[::2])

# Reverse The List -> step=-1
print("Reverse The List:", numbers[::-1])
```

4. Modifying Lists

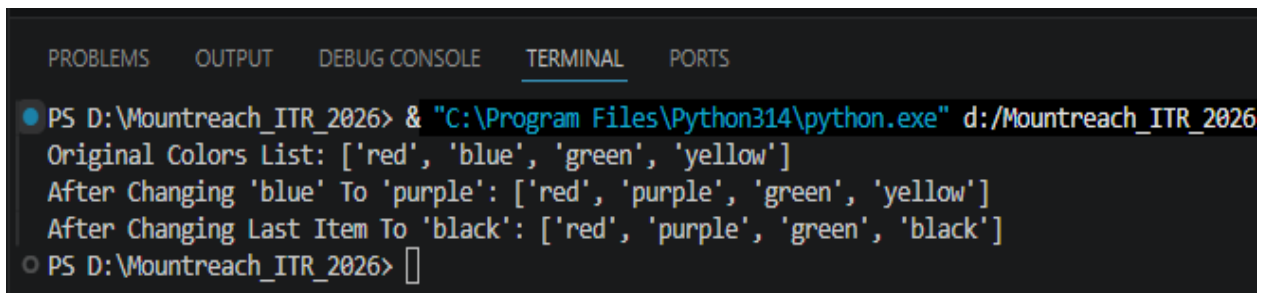
Create a list: `colors = ["red", "blue", "green", "yellow"]`

Perform the following:

- Change "blue" to "purple" using indexing.
- Change the last item to "black".

Print the list after each change.

Output:



```
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" d:/Mountreach_ITR_2026
Original Colors List: ['red', 'blue', 'green', 'yellow']
After Changing 'blue' To 'purple': ['red', 'purple', 'green', 'yellow']
After Changing Last Item To 'black': ['red', 'purple', 'green', 'black']
PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
colors = ["red", "blue", "green", "yellow"]
print("Original Colors List:", colors)

# Change "blue" To "purple" Using Its Index
colors[1] = "purple"
print("After Changing 'blue' To 'purple':", colors)

# Change The Last Item To "black"
colors[-1] = "black"
print("After Changing Last Item To 'black':", colors)
```

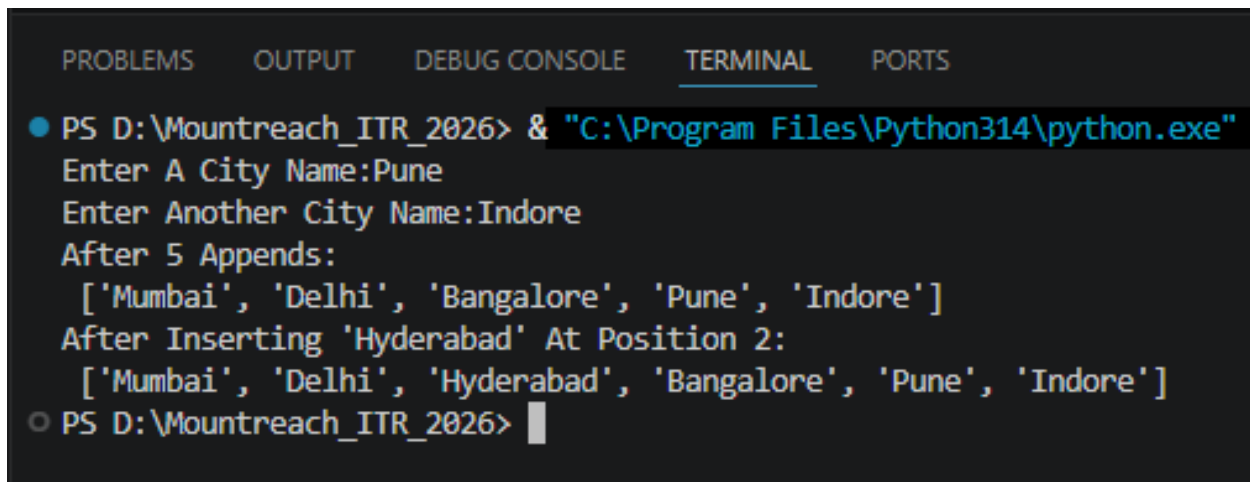
5. append() & insert()

Write a program that starts with an empty list.

- Use append() to add 5 different city names one by one.
- Use insert() to add one more city at position 2.
- Print the final list.

Take user input for at least 2 cities.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A City Name:Pune
Enter Another City Name:Indore
After 5 Appends:
['Mumbai', 'Delhi', 'Bangalore', 'Pune', 'Indore']
After Inserting 'Hyderabad' At Position 2:
['Mumbai', 'Delhi', 'Hyderabad', 'Bangalore', 'Pune', 'Indore']
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
# Empty List
cities = []

# append() Adds The Item To The End Of The List
cities.append("Mumbai")
cities.append("Delhi")
cities.append("Bangalore")

# Take User Input For Two More Cities And Append Them To The List
city1 = input("Enter A City Name:")
city2 = input("Enter Another City Name:")

cities.append(city1)
cities.append(city2)

print("After 5 Appends:\n", cities)

# insert(index, value) Places The Item At The Given Index
cities.insert(2, "Hyderabad")
print("After Inserting 'Hyderabad' At Position 2:\n", cities)
```

6. remove(), pop(), & clear()

Create a list: `items = [10, 20, 30, 20, 40, 50]`

Perform and print after each operation:

- Remove the first occurrence of 20 using `remove()`.
- Remove the item at index 3 using `pop()` and print the removed value.
- Remove the last item using `pop()`.
- Clear the entire list using `clear()`.

Add comments explaining the difference between `remove()` and `pop()`.

Output:

```
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original: [10, 20, 30, 20, 40, 50]
After Remove(20): [10, 30, 20, 40, 50]
After pop(3) Removed Value: 40 List Now: [10, 30, 20, 50]
After pop(): [10, 30, 20]
After clear(): []
PS D:\Mountreach_ITR_2026>
```

Program Code:

```
items = [10, 20, 30, 20, 40, 50]
print("Original:", items)

# remove(value) -> Removes The First Occurrence Of The Given Value
items.remove(20)
print("After Remove(20):", items)

# pop(index) -> Removes & Returns The Item At The Given Index
removed = items.pop(3)
print("After pop(3) Removed Value:", removed, "List Now:", items)

# pop() With No Argument Removes & Returns The Last Item
items.pop()
print("After pop():", items)

# clear() -> Removes All Items, Leaving An Empty List
items.clear()
```



```
print("After clear():", items)
```

```
# Difference:
```

```
# remove() -> You Specify The Value To Delete (First Match Only)
```

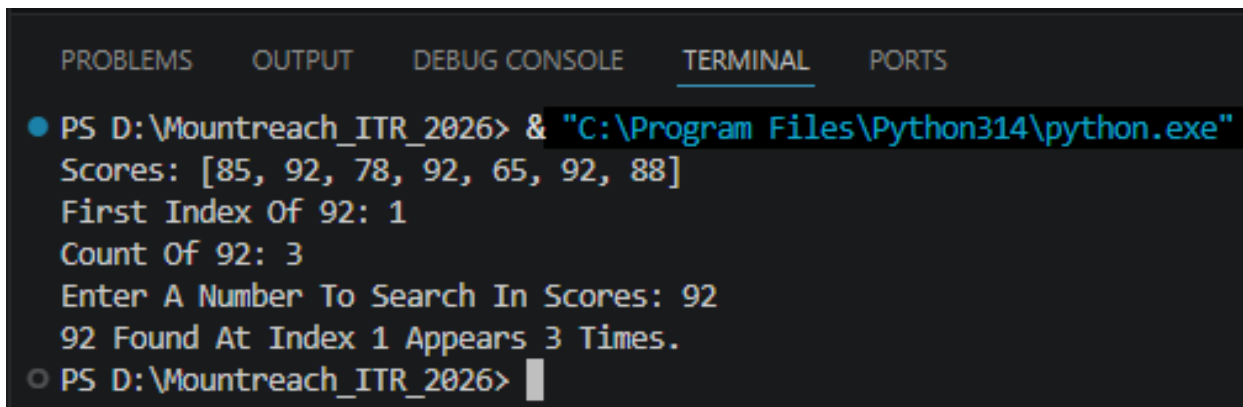
```
# pop() -> You Specify The Index To Delete
```

7. index() & count()

Create a list: scores = [85, 92, 78, 92, 65, 92, 88]

- Find and print the index of the first occurrence of 92.
- Count and print how many times 92 appears.
- Take a number as input from the user and check if it exists in the list. If yes, print its index and count.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Scores: [85, 92, 78, 92, 65, 92, 88]
First Index Of 92: 1
Count Of 92: 3
Enter A Number To Search In Scores: 92
92 Found At Index 1 Appears 3 Times.
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
scores = [85, 92, 78, 92, 65, 92, 88]
print("Scores:", scores)

# index() Returns The Position Of The First Occurrence
first_index = scores.index(92)
print("First Index Of 92:", first_index)

# count() Returns How Many Times The Value Appears
count_92 = scores.count(92)
print("Count Of 92:", count_92)

# User Input Check
num = int(input("Enter A Number To Search In Scores: "))
if num in scores:
    print(num, "Found At Index", scores.index(num), "Appears", scores.count(num),
          "Times.")
else:
    print(num, "Not Found In The List.")
```

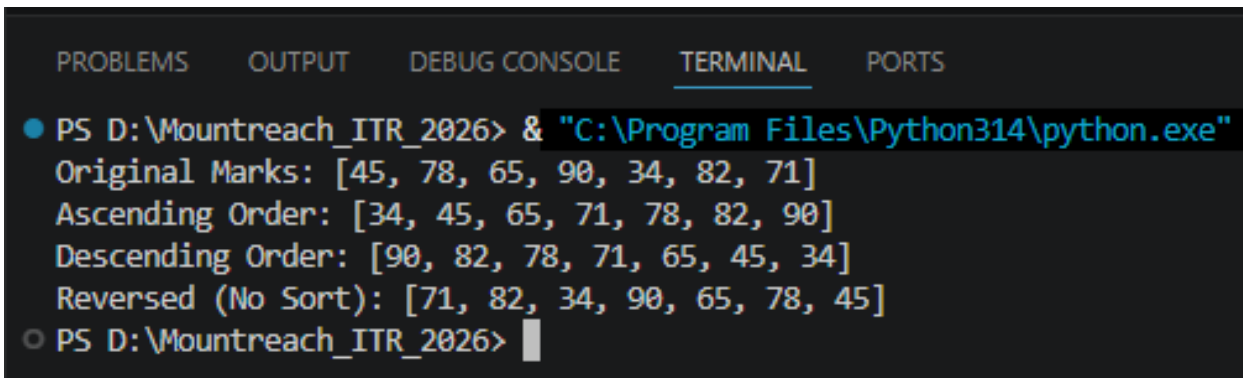
8. sort() & reverse()

Create a list: marks = [45, 78, 65, 90, 34, 82, 71]

- Sort the list in ascending order and print it.
- Sort the list in descending order and print it.
- Reverse the original list (without sorting) and print it.

Add comments on the difference between sort() and reverse().

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original Marks: [45, 78, 65, 90, 34, 82, 71]
Ascending Order: [34, 45, 65, 71, 78, 82, 90]
Descending Order: [90, 82, 78, 71, 65, 45, 34]
Reversed (No Sort): [71, 82, 34, 90, 65, 78, 45]
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
marks = [45, 78, 65, 90, 34, 82, 71]
print("Original Marks:", marks)

# sort() Rearranges Elements In Ascending Order By Default
marks.sort()
print("Ascending Order:", marks)

# sort(reverse=True) Rearranges Elements In Descending Order
marks.sort(reverse=True)
print("Descending Order:", marks)

# Reset To Original, Then Reverse()
marks = [45, 78, 65, 90, 34, 82, 71]
marks.reverse()
print("Reversed (No Sort):", marks)

# Difference:
#   sort() -> Reorders Elements By Value (Ascending or Descending)
#   reverse() -> Simply Flips The Current Order, Regardless Of Values
```

9. extend() vs append()

Create two lists:

```
list1 = [1, 2, 3]
```

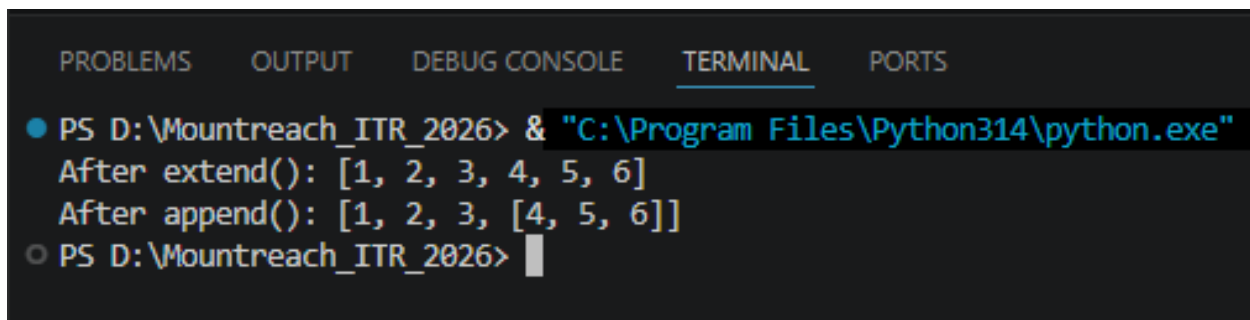
```
list2 = [4, 5, 6]
```

Demonstrate:

- Using extend() to add list2 to list1.
- Using append() to add list2 to a copy of list1.

Print both results and explain the difference in comments.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  After extend(): [1, 2, 3, 4, 5, 6]
  After append(): [1, 2, 3, [4, 5, 6]]
○ PS D:\Mountreach_ITR_2026>
```

Program Code:

```
list1 = [1, 2, 3]
list2 = [4, 5, 6]

# extend() Unpacks list2 & Adds Each Element Individually To list1
list1.extend(list2)
print("After extend():", list1)

# append() Adds list2 As A Single Nested Element
list1_copy = [1, 2, 3]
list1_copy.append(list2)
print("After append():", list1_copy)

# Difference:
# extend() -> Merges The Second List Into The First (Flat)
# append() -> Adds The Second List As One Element (Nested)
```

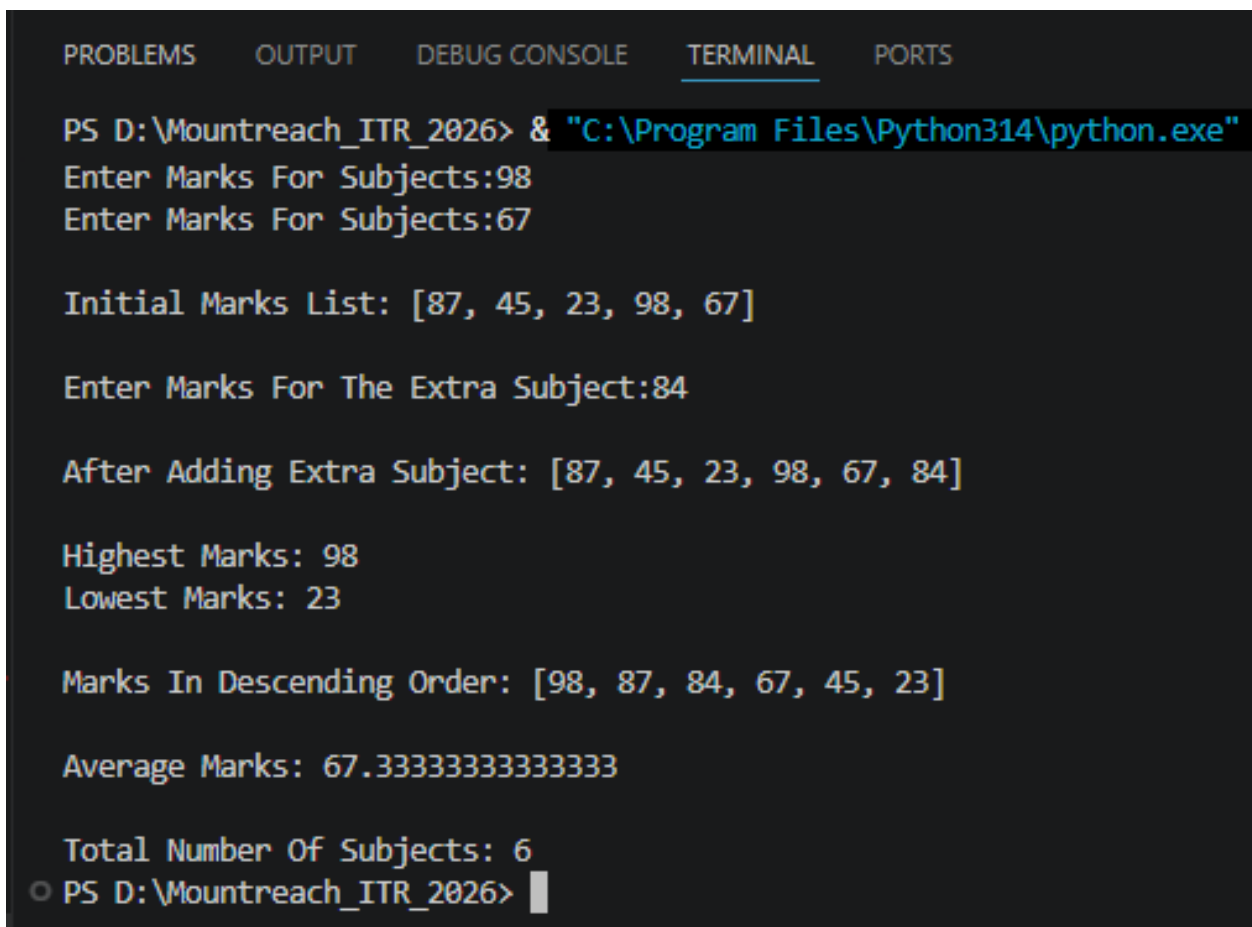
10. Mini Project - Combined Concepts

Create a **Student Marks Manager** program using lists:

- Take 5 subject marks as input from the user and store them in a list.
- Print the list.
- Add one more subject mark using `append()`.
- Print the highest and lowest marks (using built-in functions).
- Sort the marks in descending order.
- Calculate and print the average marks.
- Use `len()` to show total number of subjects.

Use proper comments, meaningful variable names, and control structures where needed.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter Marks For Subjects:98
Enter Marks For Subjects:67

Initial Marks List: [87, 45, 23, 98, 67]

Enter Marks For The Extra Subject:84

After Adding Extra Subject: [87, 45, 23, 98, 67, 84]

Highest Marks: 98
Lowest Marks: 23

Marks In Descending Order: [98, 87, 84, 67, 45, 23]

Average Marks: 67.33333333333333

Total Number Of Subjects: 6
PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
print("*** Student Marks Manager ***")

subject_marks = []

# Collect Marks For 5 Subjects From The User
for i in range(1, 6):
    mark = int(input("Enter Marks For Subjects:"))
    subject_marks.append(mark)

print("\nInitial Marks List:", subject_marks)

# Add A 6th Subject Mark
extra_mark = int(input("\nEnter Marks For The Extra Subject:"))
subject_marks.append(extra_mark)
print("\nAfter Adding Extra Subject:", subject_marks)

# Built-in max() & min() To Find Highest & Lowest Marks
print("\nHighest Marks:", max(subject_marks))
print("Lowest Marks:", min(subject_marks))

# Sort In Descending Order
subject_marks.sort(reverse=True)
print("\nMarks In Descending Order:", subject_marks)

# Average = Total / Number Of Subjects
average = sum(subject_marks) / len(subject_marks)
print("\nAverage Marks:", average)

# len() Gives The Total Count
print("\nTotal Number Of Subjects:", len(subject_marks))
```